Chapter 11, Problem 46P

Problem

For the following voltage and current phasors, calculate the complex power, apparent power, real power, and reactive power. Specify whether the pf is leading or lagging.

(a) $\mathbf{V} = 220\underline{/30^\circ}$ V rms, $I = 0.5\underline{/60^\circ}$ A rms

(b) $\mathbf{V} = 250\underline{/-10^\circ}$ V rms,
$\mathbf{I} = 6.2/-25^\circ$ A rms

(c) $\mathbf{V} = 120\underline{/0^\circ}$ V rms, $\mathbf{I} = 2.4\underline{/-15^\circ}$ A rms

(d) $\mathbf{V} = 160\underline{/45^\circ}$ V rms, $\mathbf{I} = 8.5\underline{/90^\circ}$ A rms

Step-by-step solution

Step 1 of 14

(a)

Write the phasor values of voltage and current.

$$\begin{aligned}\mathbf{V} &= V_{\text{rms}}\angle\theta_v \\ &= 220\angle 30^\circ \text{ V rms}\end{aligned}$$

$$\begin{aligned}\mathbf{I} &= I_{\text{rms}}\angle\theta_i \\ &= 0.5\angle 60^\circ \text{ A rms}\end{aligned}$$

Determine the complex power.

$$\begin{aligned}\mathbf{S} &= \mathbf{V}_{\text{rms}}\mathbf{I}^*_{\text{rms}} \\ &= (220\angle 30^\circ)(0.5\angle 60^\circ)^* \\ &= (220\angle 30^\circ)(0.5\angle -60^\circ) \\ &= 110\angle -30^\circ \text{ VA}\end{aligned}$$

$$= 95.26 - j55 \text{ VA}$$

Therefore, the complex power is $\boxed{110\angle -30^\circ \text{ VA}}$.

Determine the apparent power.

$$\begin{aligned}S &= |\mathbf{S}| \\ &= |\mathbf{V}_{\text{rms}}||\mathbf{I}_{\text{rms}}| \\ &= (220)(0.5) \\ &= 110 \text{ VA}\end{aligned}$$

Therefore, the apparent power is $\boxed{110 \text{ VA}}$.

Step 2 of 14

Determine the real power.

$$\begin{aligned} P &= \operatorname{Re}(\mathbf{S}) \\ &= S\cos(\theta_v - \theta_i) \\ &= 110\cos(30° - 60°) \\ &= 95.26 \text{ W} \end{aligned}$$

Therefore, the real power is $\boxed{95.26 \text{ W}}$.

Determine the reactive power.

$$\begin{aligned} Q &= \operatorname{Im}(\mathbf{S}) \\ &= S\sin(\theta_v - \theta_i) \\ &= 110\sin(30° - 60°) \\ &= -55 \text{ VAR} \end{aligned}$$

Therefore, the reactive power is $\boxed{-55 \text{ VAR}}$.

Step 3 of 14

Determine the power factor.

$$\begin{aligned} \text{pf} &= \cos(\theta_v - \theta_i) \\ &= \cos(30° - 60°) \\ &= 0.866 \end{aligned}$$

Since the reactive power is negative, the power factor is **0.866 leading**.

Step 4 of 14

(b)

Write the phasor values of voltage and current.

$$\begin{aligned} \mathbf{V} &= V_{\text{rms}}\angle\theta_v \\ &= 250\angle -10° \text{ V rms} \end{aligned}$$

$$\begin{aligned} \mathbf{I} &= I_{\text{rms}}\angle\theta_i \\ &= 6.2\angle -25° \text{ A rms} \end{aligned}$$

Determine the complex power.

$$\begin{aligned} \mathbf{S} &= \mathbf{V}_{\text{rms}}\mathbf{I}^*_{\text{rms}} \\ &= (250\angle -10°)(6.2\angle -25°)^* \\ &= (250\angle -10°)(6.2\angle 25°) \\ &= 1550\angle 15° \text{ VA} \end{aligned}$$

$$= 1497.18 + j401.17 \text{ VA}$$

Therefore, the complex power is $\boxed{1550\angle 15° \text{ VA}}$.

Determine the apparent power.

$$\begin{aligned} S &= |\mathbf{S}| \\ &= |\mathbf{V}_{\text{rms}}||\mathbf{I}_{\text{rms}}| \\ &= (250)(6.2) \\ &= 1550 \text{ VA} \end{aligned}$$

Therefore, the apparent power is $\boxed{1550 \text{ VA}}$.

Step 5 of 14

Determine the real power.

$$\begin{aligned} P &= \text{Re}(\mathbf{S}) \\ &= S\cos(\theta_v - \theta_i) \\ &= 1550\cos(-10° + 25°) \\ &= 1497.18 \text{ W} \end{aligned}$$

Therefore, the real power is $\boxed{1497.18 \text{ W}}$.

Step 6 of 14

Determine the reactive power.

$$\begin{aligned} Q &= \text{Im}(\mathbf{S}) \\ &= S\sin(\theta_v - \theta_i) \\ &= 1550\sin(-10° + 25°) \\ &= 401.17 \text{ VAR} \end{aligned}$$

Therefore, the reactive power is $\boxed{401.17 \text{ VAR}}$.

Step 7 of 14

Determine the power factor.

$$\begin{aligned} \text{pf} &= \cos(\theta_v - \theta_i) \\ &= \cos(-10° + 25°) \\ &= \cos(15°) \\ &= 0.966 \end{aligned}$$

Since the reactive power is positive, the power factor is **0.966 lagging**.

Step 8 of 14

(c)

Write the phasor values of voltage and current.

$$\mathbf{V} = V_{\text{rms}}\angle\theta_v$$
$$= 120\angle 0° \text{ V rms}$$
$$\mathbf{I} = I_{\text{rms}}\angle\theta_i$$
$$= 2.4\angle -15° \text{ A rms}$$

Determine the complex power.

$$\mathbf{S} = \mathbf{V}_{\text{rms}}\mathbf{I}^*_{\text{rms}}$$
$$= (120\angle 0°)(2.4\angle -15°)^*$$
$$= (120\angle 0°)(2.4\angle 15°)$$
$$= 288\angle 15° \text{ VA}$$
$$= 278.19 + j74.54 \text{ VA}$$

Therefore, the complex power is $\boxed{288\angle 15° \text{ VA}}$.

Determine the apparent power.

$$S = |\mathbf{S}|$$
$$= |\mathbf{V}_{\text{rms}}||\mathbf{I}_{\text{rms}}|$$
$$= (120)(2.4)$$
$$= 288 \text{ VA}$$

Therefore, the apparent power is $\boxed{288 \text{ VA}}$.

Step 9 of 14

Determine the real power.

$$P = \text{Re}(\mathbf{S})$$
$$= S\cos(\theta_v - \theta_i)$$
$$= 288\cos(0° + 15°)$$
$$= 278.19 \text{ W}$$

Therefore, the real power is $\boxed{278.19 \text{ W}}$.

Determine the reactive power.

$$Q = \text{Im}(\mathbf{S})$$
$$= S\sin(\theta_v - \theta_i)$$
$$= 288\sin(0° + 15°)$$
$$= 74.54 \text{ VAR}$$

Therefore, the reactive power is $\boxed{74.54 \text{ VAR}}$.

Step 10 of 14

Determine the power factor.

$$\text{pf} = \cos(\theta_v - \theta_i)$$
$$= \cos(0° + 15°)$$
$$= \cos(15°)$$
$$= 0.966$$

Since the reactive power is positive, the power factor is **0.966 lagging**.

Step 11 of 14

(d)

Write the phasor values of voltage and current.

$$\begin{aligned}\mathbf{V} &= V_{\text{rms}}\angle\theta_v \\ &= 160\angle 45^\circ \text{ V rms}\end{aligned}$$

$$\begin{aligned}\mathbf{I} &= I_{\text{rms}}\angle\theta_i \\ &= 8.5\angle 90^\circ \text{ A rms}\end{aligned}$$

Determine the complex power.

$$\begin{aligned}\mathbf{S} &= \mathbf{V}_{\text{rms}}\mathbf{I}_{\text{rms}}^* \\ &= (160\angle 45^\circ)(8.5\angle 90^\circ)^* \\ &= (160\angle 45^\circ)(8.5\angle -90^\circ) \\ &= 1360\angle -45^\circ \text{ VA}\end{aligned}$$

$$= 961.66 - j961.66 \text{ VA}$$

Therefore, the complex power is $\boxed{1360\angle -45^\circ \text{ VA}}$.

Step 12 of 14

Determine the apparent power.

$$\begin{aligned}S &= |\mathbf{S}| \\ &= |\mathbf{V}_{\text{rms}}||\mathbf{I}_{\text{rms}}| \\ &= (160)(8.5) \\ &= 1360 \text{ VA}\end{aligned}$$

Therefore, the apparent power is $\boxed{1360 \text{ VA}}$.

Step 13 of 14

Determine the real power.

$$\begin{aligned}P &= \text{Re}(\mathbf{S}) \\ &= S\cos(\theta_v - \theta_i) \\ &= 1360\cos(45^\circ - 90^\circ) \\ &= 961.66 \text{ W}\end{aligned}$$

Therefore, the real power is $\boxed{961.66 \text{ W}}$.

Determine the reactive power.

$$\begin{aligned}Q &= \text{Im}(\mathbf{S}) \\ &= S\sin(\theta_v - \theta_i) \\ &= 1360\sin(45^\circ - 90^\circ) \\ &= -961.66 \text{ VAR}\end{aligned}$$

Therefore, the reactive power is $\boxed{-961.66 \text{ VAR}}$.

Step 14 of 14

Determine the power factor.

$$\begin{aligned}\text{pf} &= \cos(\theta_v - \theta_i) \\ &= \cos(45^\circ - 90^\circ) \\ &= 0.707\end{aligned}$$

Since the reactive power is negative, the power factor is **0.707 leading**.